TANUSH THAKRAN

AI / ML Engineer

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SUMMARY

Machine learning engineer who ships models with evidence behind them: leakage-safe evaluation, controlled ablations, and metrics reported against honest baselines. Experience spans learning-to-rank on multi-million-row datasets, applied ML with scikit-learn, and LLM/RAG systems, owning the full path from data extraction through deployment.

TECHNICAL SKILLS

Machine Learning: LightGBM, scikit-learn, Feature Engineering, Cross-Validation, PyTorch, TensorFlow

LLM & Retrieval: RAG, Embeddings, Vector Search, Prompt Engineering, LLM Evaluation, Groq API, Whisper, Ollama

Data Engineering: Python, Pandas, NumPy, SQL, PyArrow/Parquet, Multiprocessing

Web & Infra: React.js, Node.js, FastAPI, PostgreSQL, Docker, Streamlit, Git, Flutter

MACHINE LEARNING & DATA PROJECTS

Pokémon TCG Move Prediction — Learning-to-Rank

Python, LightGBM, PyArrow | *github.com/Tanush1206/pokemon-tcg-move-prediction*

- Framed move selection as learning-to-rank (not classification) and trained an LGBMRanker with a LambdaRank objective on 3.35M rows and 83 features, reaching 57.4% top-1 accuracy vs. a 21.7% random baseline (2.65x lift).
- Ran a 7-run ablation study with episode-level, leakage-safe splits; largest single gain (+0.054 accuracy) came from reverse-engineering undocumented card-identity indices.

Customer Churn Prediction — Deployed ML App

Python, scikit-learn, Streamlit | *github.com/Tanush1206/Customer-Churn-Prediction*

- Built an end-to-end telecom churn pipeline (leakage-safe ColumnTransformer, Logistic Regression / Random Forest / Gradient Boosting compared via 5-fold CV), deployed as an interactive Streamlit app; selected a 0.30 decision threshold via cross-validation to prioritize recall, reaching 75.1% recall and 84.1% ROC-AUC on held-out data.

RAG-Based AI — Offline Retrieval-Augmented QA Pipeline

Python, Whisper, Ollama | *github.com/Tanush1206/rag_based_ai*

- Built a fully offline RAG pipeline (Whisper transcription, local bge-m3 embeddings, cosine-similarity retrieval, deepseek-r1 via Ollama) that grounds every answer in a cited video, title, and timestamp — zero external API calls.

LLM Utility Lab

Python, Groq API | *github.com/Tanush1206/llm-utility-lab*

- Built a modular LLM application — separate modules for client communication, prompting, summarization, context-aware Q&A, and evaluation — with per-request token tracking and a dedicated evaluation runner backed by automated pytest coverage.

Superstore Profitability Analysis — SQL & Power BI

SQL, PostgreSQL, Power BI, DAX | *github.com/Tanush1206/superstore-powerbi-analysis*

- Bucketed 9,994 transactions ($2.3M revenue) by discount band in SQL — margin turns negative past 25% discount, reaching -77% beyond 40% — and shipped an interactive Power BI dashboard (DAX, cohort/RFM segmentation) recommending ~$35K in recoverable annual profit.

VideoCaptionMaker — Offline AI Captioning & Translation App

Python, FastAPI, Celery, Next.js, Docker | *github.com/Tanush1206/Video-Caption-Maker*

- Built a private, offline captioning app: local faster-whisper transcription and M2M100 translation (9 languages incl. Hinglish) on CPU or NVIDIA GPU, with hardware-aware model selection and semantic transcript search.
- Shipped a one-command Docker installer with SHA-256-verified updates (cut from ~2.2 GB to ~6 MB), crash recovery, and 350+ tests.

EXPERIENCE

App Developer Intern — CYPARTA Dec 2025 – Jan 2026

Cairo, Egypt (Remote) | *Flutter, Firebase*

- Built the Driver & Medic mobile app end-to-end, handling real-time GPS telemetry and state consistency across concurrent users for live logistics coordination.
- Implemented full trip-lifecycle workflows and post-trip EMS reporting modules for data integrity.

Junior Manager — AIESEC Oct 2025 – Mar 2026

Bengaluru, India | *Managed Global Volunteer operations, tracked departmental KPIs, led cross-functional teams.*

EDUCATION

B.Sc. Computer Science — BITS Pilani 2024 – 2028

CGPA: 7.64 | *Coursework: Algorithms, Data Structures, Database Systems, Operating Systems, Computer Networks, System Design, Statistics, Deep Learning*

CERTIFICATIONS

Machine Learning: Advanced Learning Algorithms — Stanford / DeepLearning.AI • Supervised ML: Regression & Classification — Stanford / DeepLearning.AI • Intermediate Machine Learning — Kaggle

LLM & GenAI: LangChain: Chat with Your Data (LangChain, DeepLearning.AI) • Building Systems with the ChatGPT API (OpenAI, DeepLearning.AI) • ChatGPT Prompt Engineering for Developers (OpenAI, DeepLearning.AI) • AI Fluency: Framework & Foundations (Anthropic)